

Exactly. **We are not building Unity geometry first.** The first thing we are building is the **canonical Phantom Q Headquarters design specification**. Once that is internally consistent, the Unity greybox becomes implementation rather than experimentation.

And we should work through A1–A10 in a **looping validation process**, not simply finish A1, forget it, and move down a checklist.

What we build first

Product 0 — Phantom Q HQ Canonical Specification

Before Unity construction, we produce one master design containing:

```
PHANTOM Q HQ — CANONICAL SPECIFICATION
|
|— MASTER FLOOR PLAN
|
|— DEPARTMENT RELATIONSHIPS
|
|— CAMPAIGN ROUTES
|   └─ Prologue → Incident 06
|
|— CHARACTER ROUTES
|   └─ Player / Alice / Bob / Eve / Red Team
|
|— CLEARANCE MAP
|
|— INTERACTION MAP
|
|— CAMERA / HANDLING RULES
|
|— ROOM PURPOSES
|
|— FUTURE EXPANSION
|   └─ Engineering / Quantum / Physical QKD
|
|— UNITY IMPLEMENTATION PARTITION
```

Once that specification survives our checks, **then we build the entire HQ greybox in Unity.**

How the looping process works

We are not going to do this:

```
A1 ✓
forget about it
```

A2 ✓
forget about it

A3 ✓
...

We do this:

A1 BUILDING MEANING

↓

A2 SPATIAL HIERARCHY

↓

Does A2 still support A1?

↓

YES → continue

NO → correct A1/A2

↓

A3 ADJACENCY

↓

Check against A1 + A2

↓

A4 FLOOR PLAN

↓

Run A1+A2+A3 AGAIN

↓

A5 CAMPAIGN ROUTES

↓

Does Prologue → 06 work physically?

↓

A6 CHARACTER ROUTES

↓

Do Alice/Eve/Red Team routes still make sense?

↓

A7 CLEARANCE

↓

Does progression work through the building?

↓

A8 INTERACTIONS

↓

Do rooms now have meaningful gameplay?

↓

A9 CAMERA/HANDLING

↓

Can we actually PLAY this layout?

↓

A10 UNITY PARTITION

↓

Can we build this cleanly?

↓

FINAL SYSTEM LOOP

A1 → A10

↓

CANONICAL HQ LOCK

That is the process I would trust before we start serious Unity construction.

So we start now with A1 — Building Meaning

This sounds abstract, but it actually controls everything that comes later.

The first question is:

What is Phantom Q Headquarters supposed to communicate through its physical architecture?

My proposed answer is:

Phantom Q Headquarters is an operational engineering and security facility in which physical access reflects increasing technical competence, operational trust and responsibility.

That gives us three architectural ideas simultaneously:

TECHNICAL CAPABILITY

+

SECURITY TRUST

+

PLAYER PROGRESSION

The building should physically represent all three.

A1.1 — It must feel like a real organisation

We should avoid building something that feels like:

ROOM 1 = tutorial

ROOM 2 = symmetric lesson

ROOM 3 = asymmetric lesson

ROOM 4 = QKD lesson

That would feel like an educational theme park.

Instead, departments exist because **Phantom Q as an organisation needs them.**

For example:

Central Operations

Exists because Phantom Q operates systems and missions.

Not because:

“The Prologue needs a room.”

SOC

Exists because Phantom Q monitors and investigates security events.

Incident 04 and 06 then naturally use it.

Cryptography Laboratory

Exists because Phantom Q researches/tests secure information protection.

Incidents 02, 03 and 05 naturally use it.

Red Team Operations

Exists because authorised adversarial/security testing is part of Phantom Q's operational capability.

Incident 06 eventually gives the learner access.

Quantum Engineering

Exists because quantum technologies are part of Phantom Q's wider engineering mission.

It is not arbitrarily added later because the game reaches “the QKD level”.

This distinction matters enormously.

A1.2 — The building represents increasing trust

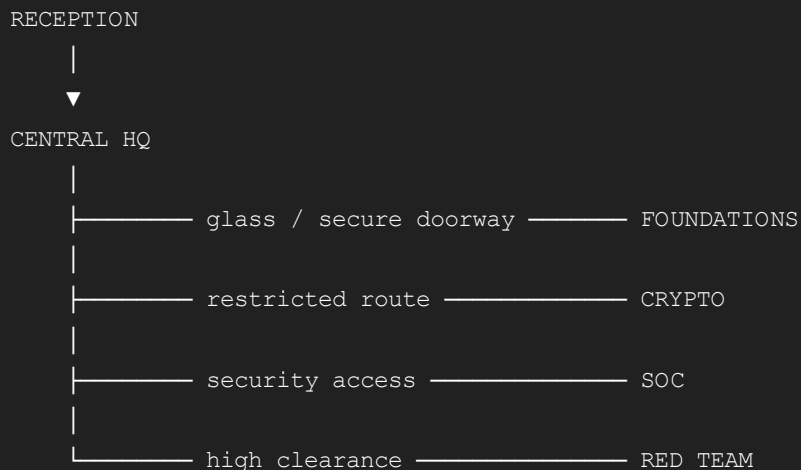
At the beginning, the trainee should feel:

```
I AM INSIDE PHANTOM Q  
  
BUT  
  
I DO NOT YET HAVE ACCESS  
TO PHANTOM Q'S FULL CAPABILITY.
```

Therefore the architecture must allow the player to **see more than they can initially access**.

That's important.

For example:



The player can see staff moving somewhere they cannot yet enter.

Later:

```
ACCESS GRANTED
```

and that same door opens.

That is much stronger progression than a menu saying:

```
| Level unlocked.
```

A1.3 — There should be a security-depth progression

The Headquarters should transition from:

PUBLIC / CONTROLLED
↓
NORMAL OPERATIONS
↓
SPECIALIST OPERATIONS
↓
SECURITY OPERATIONS
↓
ADVANCED ENGINEERING
↓
HIGH-TRUST SYSTEMS

I would therefore define five conceptual security bands.

Band 1 — Identity / Entry

ENTRANCE
RECEPTION
IDENTITY / ACCESS CONTROL

Purpose:

Who are you, and should you enter?

This fits the Prologue immediately.

Band 2 — Operational Core

CENTRAL OPERATIONS
COMMUNICATIONS

Purpose:

Day-to-day Phantom Q work happens here.

This is where the player initially becomes operational.

Band 3 — Specialist Capability

TRAINING & FOUNDATIONS
CRYPTOGRAPHY LAB
ENGINEERING

Purpose:

Learn/build specialist capability.

These are not necessarily all equally restricted, but they represent specialist technical functions.

Band 4 — Security Response

SOC

RED TEAM OPERATIONS

Purpose:

SOC

Detect / investigate / correlate

RED TEAM

Authorised active security response

They should be related spatially but separately controlled.

Band 5 — Advanced / High Trust

QUANTUM WING

ADVANCED COMPUTE

SECURE CORE

Purpose:

Phantom Q's most advanced and restricted capabilities.

That gives the future game somewhere to grow.

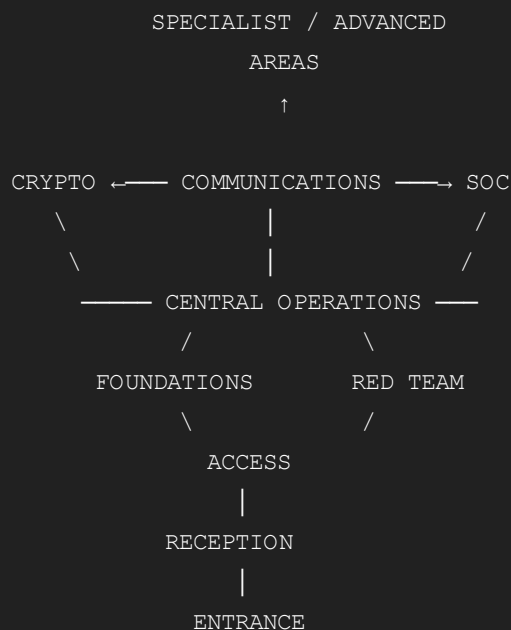
A1.4 — Central Operations is the organisational heart

Security depth does **not** mean everything should be arranged in one long corridor from low clearance to high clearance.

That would make the map boring.

Instead, Central Operations should be the **hub**.

Conceptually:



Again, that's **relationship logic**, not the final floor-plan geometry yet.

Central Operations remains the place the learner repeatedly comes back to.

A1.5 — The building should tell the campaign story

The physical journey from Prologue through Incident 06 should represent progression.

Prologue

```
OUTSIDE
↓
IDENTITY
↓
CENTRAL OPERATIONS
```

The learner enters the organisation.

Incident 01

```
CENTRAL OPS
↓
FOUNDATIONS
```

The learner learns how Phantom Q reasons.

Incidents 02–03

CENTRAL OPS
↓
CRYPTO
↔
COMMUNICATIONS

The learner develops technical protection capability.

Incident 04

CENTRAL OPS
↓
COMMUNICATIONS
↓
SOC

The world shifts from normal operations into security response.

Incident 05

SOC
↓
CRYPTO
↓
COMMUNICATIONS
↓
RED TEAM STANDBY

The learner develops enough trust infrastructure for recovery operations.

Incident 06

SOC
↔
RED TEAM
↔
HQ AUTHORITY
↓
RECOVERY

↓

RESTORATION

↓

CENTRAL OPERATIONS

The learner returns to the place where the campaign essentially started.

But now they are not the same trainee.

That's important spatial storytelling.

A1.6 — The building should change state, not change building

The HQ itself stays persistent.

For example, Central Operations:

Beginning

STATUS: NORMAL

routine work

staff moving

standard displays

PQ-001

STATUS: INCIDENT REVIEW

local warning

security activity

Incident 04

STATUS: RESTRICTED

SOC escalation

A7 suspended

financial impact

Incident 06

STATUS: RECOVERY OPERATION

Red Team status
recovery progress

End

STATUS: RESTORED

PQ-004 closed
services operational
actor unresolved

That is environmental storytelling without building another scene.

A1.7 — Departments must have operational relationships

This will become critical when we do A3 adjacency.

We should already establish the **reason relationships exist**.

Crypto ↔ Communications

CRYPTO
protects information

COMMUNICATIONS
moves information

Natural relationship.

Communications ↔ SOC

COMMUNICATIONS
generates sessions / traffic

SOC
monitors / investigates those activities

Natural relationship.

SOC ↔ Red Team

SOC
finds / analyses

RED TEAM
acts under authority

Natural relationship.

Engineering ↔ Quantum

ENGINEERING
hardware / embedded / integration

QUANTUM
specialist optical/QKD systems

Natural relationship.

Advanced Compute ↔ Quantum / Secure Core

Potential future relationship:

ADVANCED COMPUTE
simulation / high-performance processing

QUANTUM
advanced security technology

SECURE CORE
highest-trust operational infrastructure

That gives us future expansion without distorting Release 1.

A1.8 — The building should support different kinds of gameplay

We don't want every room to become:

walk in
↓

sit at another monitor

↓

read text

Spatial design should support different behaviours.

Central Operations

MOVE
SPEAK
TRANSFER
MONITOR
RESPOND

Foundations

ARRANGE
COMPARE
RECONSTRUCT
CLASSIFY

Crypto

BUILD
CONNECT
TRANSFORM
TEST

Communications

OBSERVE
SEND
TRACE PATHS
COMPARE ENDPOINTS

SOC

CORRELATE
FILTER
BUILD TIMELINE
ASSESS

Red Team

TARGET
REQUEST
VERIFY
ENGAGE
SEARCH
PRESERVE

Quantum

Later:

ALIGN
MEASURE
CONFIGURE
OBSERVE PHOTONS
CONNECT PHYSICAL HARDWARE

This means room architecture must eventually accommodate **different physical interaction styles**.

A1.9 — The Headquarters must represent Engineering too

This is important because **Phantom Q Engineers should not become merely a cybersecurity office**.

The building needs visible evidence of engineering:

- hardware benches;
- equipment areas;
- cabling;
- embedded systems;
- test equipment;
- optical/QKD facilities;
- technical displays;
- controlled hardware access.

Even during the early cyber incidents, the learner should understand:

This organisation builds systems as well as analyses them.

That will make the later physical QKD integration feel native rather than bolted on.

A1.10 — South African identity without turning it into decoration

The Headquarters should feel like an advanced technical organisation that can plausibly exist in our setting.

That can emerge through:

- diverse South African staff;
- restrained localisation;
- signage/language where appropriate;
- engineering context;
- realistic infrastructure;
- local institutional/professional cues where suitable.

But we should avoid making the architecture a collection of stereotypical visual motifs.

The primary identity remains:

PHANTOM Q ENGINEERS

A1.11 — The player should always know roughly where they are

The HQ cannot become a labyrinth.

Our design rule should be:

The player should normally be able to mentally return to Central Operations without opening a map.

Therefore:

- short wings;
- strong landmarks;
- controlled sightlines;
- consistent orientation;
- distinct room identities;
- no maze corridors.

This directly supports our fixed oblique camera.

A1.12 — Future areas must already have architectural existence

This is a major advantage of designing the whole HQ now.

We don't need to finish everything visually for Release 1, but the building should already accommodate:

ENGINEERING
QUANTUM
ADVANCED COMPUTE
SECURE CORE

Even if early versions show:

RESTRICTED ACCESS

That prevents future Acts from requiring:

“A mysterious new wing suddenly appeared.”

The world was always larger than the learner's access.

A1.13 — Physical QKD has to influence the architecture now

Even though we are not building it first, we know a physical QKD demonstrator is part of the wider system.

Therefore the Quantum/Engineering area eventually needs to accommodate:

DIGITAL QKD SIMULATION
↓
PHYSICAL HARDWARE
↓
ALICE
EVE
BOB

That means the future room cannot just be a 4 × 4 office with one computer.

It needs:

- optical bench/workbench space;
- physical hardware stations;

- learner viewing space;
- cables/control equipment;
- safety/circulation clearance;
- potentially three role stations.

That requirement belongs in A1 now because it will affect A4 room sizing later.

A1.14 — Meaningful security architecture

Doors should not just be game blockers.

Different access boundaries should reflect different reasons:

IDENTITY BOUNDARY

Who may enter HQ?

OPERATIONAL BOUNDARY

Who may use operational systems?

SPECIALIST BOUNDARY

Who may access technical laboratories?

SECURITY BOUNDARY

Who may access SOC/Red Team capability?

HIGH-TRUST BOUNDARY

Who may access Quantum/Secure Core?

So later when we implement clearance in C#, the architecture already has logic behind it.

A1.15 — What the building should make the player feel

This becomes our qualitative test.

Arrival

“This place is larger than what I currently understand.”

Foundations

“I am beginning to understand how Phantom Q operates.”

Crypto

“I now have access to real technical capability.”

SOC

“Something serious has happened.”

Red Team

“I am now trusted with actions I could not perform earlier.”

Quantum/Secure Core later

“I have reached capabilities that were visibly beyond me at the beginning.”

If the architecture supports that emotional progression, it is doing its job.

A1 — Building Meaning: Proposed Lock

So I would currently carry these **15 design rules** into A2:

1. **One persistent Headquarters.**
2. **The building is an operational engineering/security facility, not a tutorial theme park.**
3. **Central Operations is the spatial and narrative heart.**
4. **Access depth represents competence, trust and responsibility.**
5. **Future restricted areas are visible before they are accessible.**
6. **Departments exist for organisational reasons independent of the campaign.**
7. **Campaign incidents use those existing functions naturally.**
8. **Crypto, Communications, SOC and Red Team have clear but distinct relationships.**
9. **Engineering and Quantum are fundamental parts of Phantom Q's identity.**
10. **Physical QKD requirements must already influence the master architecture.**
11. **Rooms must support distinct gameplay rather than repeating generic computer interactions.**
12. **The same rooms change operational state as the campaign evolves.**
13. **The HQ remains compact, legible and non-maze-like.**
14. **The campaign begins and effectively closes through Central Operations, creating a spatial loop.**
15. **The architecture must accommodate later expansion without redesigning the whole facility.**

Now we have something useful to loop against.

The **next actual design operation is A2 — Spatial Hierarchy**: taking these principles and determining **which zones belong near the entrance, around the hub, in specialist wings, behind secondary security boundaries, and in the deepest/highest-trust part of HQ**. That is where the floor plan begins to take physical shape, but we still do not draw the final geometry until the relationships survive the loop.